Fourier Transform

→ Whenever we analyze a signal, it is alway simplier when we use frequency domain characteristics of the signal.

→ Therefore, Fourier transformation is very import tools to analyze signal.

→ Analyze of frequency domain signal is much more easy to handle compared to time domain signal.

→ We analyze Discrete signal in a Frequency domain, here we called Discrete Fourier Transform (DFT).

→ DFT is originatted from Discrete-Time Fourier Transform (DTFT)

DTFT

By definition: $X(e^{j\omega}) \sum_{0}^{\infty} x(n) e^{-j\omega n}$ [single sided]

- Here transforming n to ω. [$n \rightarrow \omega$]
  n ↓ Discrete time, ω ↓ frequency

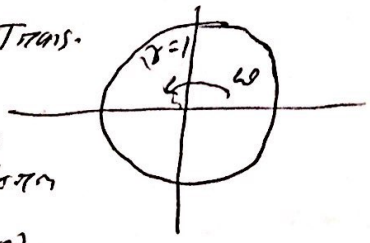
ω-Domain is similar to Z-Domain

Continuous ~~Freque~~ Frequency Trans.

$\Omega = -\infty \; +\infty$ (Non-periodic)

Discrete Frequency Transform

$\omega = 0 - 2\pi$ (Periodic)



$\omega = 0 - 2\pi$

[Time is Discrete but frequency is continuous]

$x(n) \xrightarrow{F.T.} X(e^{j\omega})$

$x(n)$ ↓ Discrete signal

$X(e^{j\omega})$ ↓ Continuous ω $(0, 2\pi)$

IDTFT

$$x(n) = \frac{1}{2\pi}\int_{0}^{2\pi} X(e^{j\omega}) e^{j\omega n} \, d\omega$$

DTFT

$$X(\omega) = \sum_{n=0}^{\infty} x(n) e^{-j\omega n}$$

Scanned with CamScanner

Discrete Fourier Transform (DFT)

$$x(n) \underset{IDFT}{\overset{DFT}{\rightleftarrows}} X(k)$$

By defn: $X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi nk}{N}}$; $k = 0, 1 \ldots N-1$

IDFT: $x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{+j\frac{2\pi nk}{N}}$; $n = 0, 1 \ldots N$

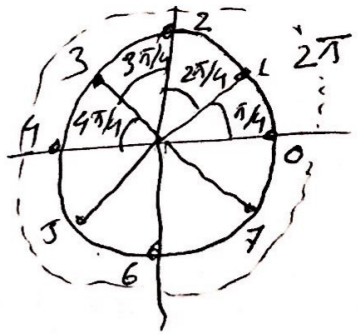
$k = 0 \ldots 7$
$(N-1)$

Relation (DTFT & DFT)

DTFT: $X(\omega) = \sum_{n=0}^{\infty} x(n)\, e^{-j\omega n}$

DFT: $X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi nk}{N}}$

$\omega = \frac{2\pi k}{N}$



Ans: 0 $(2\pi/8)$ $0(\pi/8)$ 2 $(2\pi/8)$
1 $(2\pi/8)$ 3 $(2\pi/8)$

DFT is the frequency sampled version of DTFT.
(Ω - continuous)
(ω is discrete - 0 - 2π)
(Ω is continuous)

Scanned with CamScanner

DFT Example

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\omega n} \qquad \omega = \frac{2\pi k}{N}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi n k}{N}} \quad ; \; k = 0 \cdots N-1$$

4 point DTF

$$= \sum_{n=0}^{3} x(n)\, e^{-\frac{j2\pi n k}{4}} = \sum_{n=0}^{3} x(n)\, e^{-j\frac{\pi n k}{2}}$$

$$= x(0)e^{0} + x(1)e^{-j\frac{2\pi k}{4}} + x(2)e^{-j\frac{2\pi(2)k}{4}} + x(3)e^{-j\frac{2\pi(3)k}{4}}$$

$$X(0) = x(0) + x(1) + x(2) + x(3) \qquad [e^{0} = 1]$$

$$X(1) = x(0) + x(1)e^{-j\frac{2\pi}{4}} + x(2)e^{-j\pi} + x(3)e^{-j\frac{3\pi}{2}}$$

$$X(2) = x(0) + x(1)e^{-j\pi} + x(2)e^{-j2\pi} + x(3)e^{-j\frac{3\pi}{2}}$$

$$X(3) = x(0) + x(1)e^{-j3\pi/2} + x(2)e^{-j3\pi} + x(3)e^{-j\frac{9\pi}{2}}$$

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi n k}$$

$$= \sum_{n=0}^{N-1} x(n)\, W_N^{nk}$$

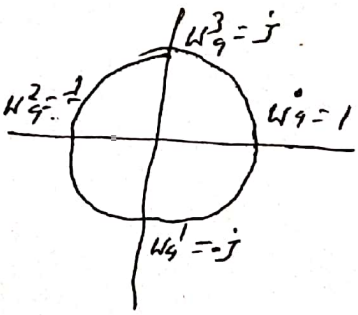
$\hookrightarrow$ Twiddle Factor

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} \text{Twiddle} \\ \text{matrix} \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

$$W_N^{1} = e^{-j\frac{2\pi(1)}{N}}$$

* Calculate DFT from given $x(n) = \{1 \;\; 2 \;\; 3 \;\; 4\}$; $N = 4$

$$X(k) = \begin{array}{c} \\ k=0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} n=0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \end{array} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$



$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1+2+3+4 \\ 1-2j-3+4j \\ 1-2+3-4 \\ 1+2j-3-4j \end{bmatrix} = \begin{bmatrix} 10 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix}$$

Scanned by CamScanner

Twiddle Factor

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad \text{--- DFT}$$

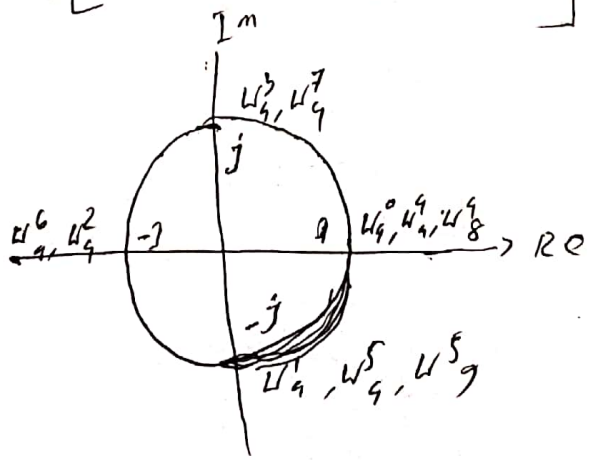
$$x(n) = \frac{1}{N} \sum_{n=0}^{N-1} X(k) W_N^{-kn} \quad \text{--- IDFT}$$

$$W_N = e^{-j2\pi/N}$$

4 point Twiddle Factor

$$W = e^{-j\left(\frac{2\pi}{4}\right)}$$

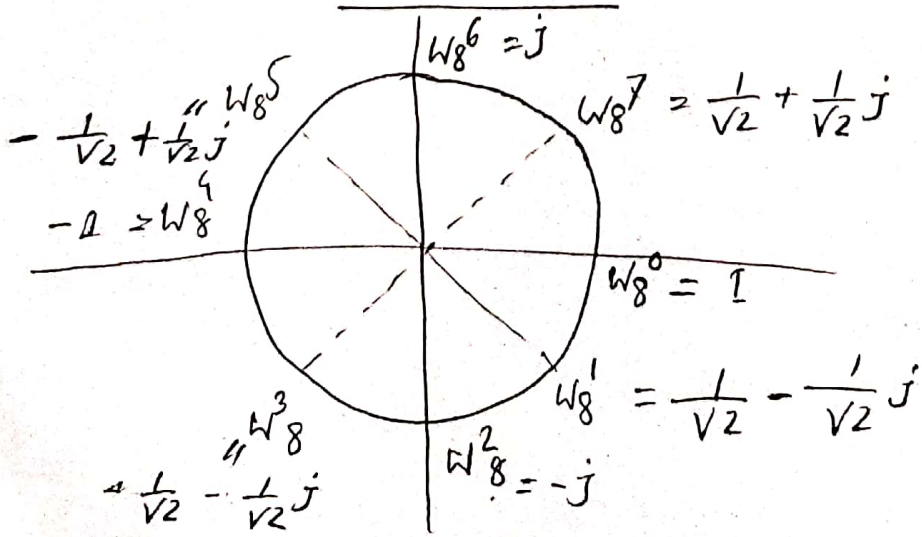
$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{array}{c} \\ k=0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} n=0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_6^4 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \end{array} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3 \end{bmatrix}$$



$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$\theta - 2\pi$$

8 point



$N = 8$

$W_N^4 = -1$

Scanned by CamScanner

Fast Fourier Transform

$x(n) = \{1 \quad 2 \quad 3 \quad 4\}, \quad N = 4$

$X(k) = \sum_{n=0}^{3} x(n) W_N^{nk} \qquad \because W_N = e^{-j\frac{2\pi}{N}}$

$X(k) = x(0)W_4^0 + x(1)W_4^k + x(2)W_4^{2k} + x(3)W_4^{3k}$

$X(0) =$

$X(1) =$

$X(2) =$

$X(3) =$

Complex multiplication $= 4 \times 4 = 16$

Complex Addition $= 3 \times 4 = 12$

Two Professor: Cooley & Tuckey developed fast computation of Fourier Transform (or) Fast Fourier Transform (FFT)

N-pt DFT

- N/2 pt
  - N/4 pt
  - N/4 pt
- N/2 pt
  - N/4 pt
  - N/4 pt.

Sub-divided & then add.

4 pt DFT → 2 pt DFT, 2 pt. DFT

8 → 4 → 2, 2; 4 → 2, 2

32, 64, 128

$\therefore N = 2^r$

r: Base (Radix)

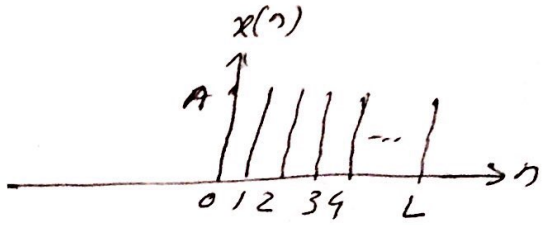
Radix 2

- DIT (Decimation in Time) $x(n)$
- DIF (Decimation in Frequency) $X(k)$

Scanned with CamScanner

Example

$$x(n) = \begin{cases} A & \text{for } 0 \le n \le L \\ 0 & \text{otherwise} \end{cases}$$



$X(\omega) = ?$

Soln: DTFT

$$X(\omega) = \sum_{n=0}^{\infty} x(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{L} x(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{L} A e^{-j\omega n}$$

$$= A \sum_{n=0}^{L} e^{-j\omega n}$$

$$= A \left[\frac{1 - e^{-j\omega(L+1)}}{1 - e^{j\omega}} \right]$$

We know,

$$\sum_{n=0}^{K} a^n = \frac{1 - a^{K+1}}{1 - a}$$

Scanned with CamScanner

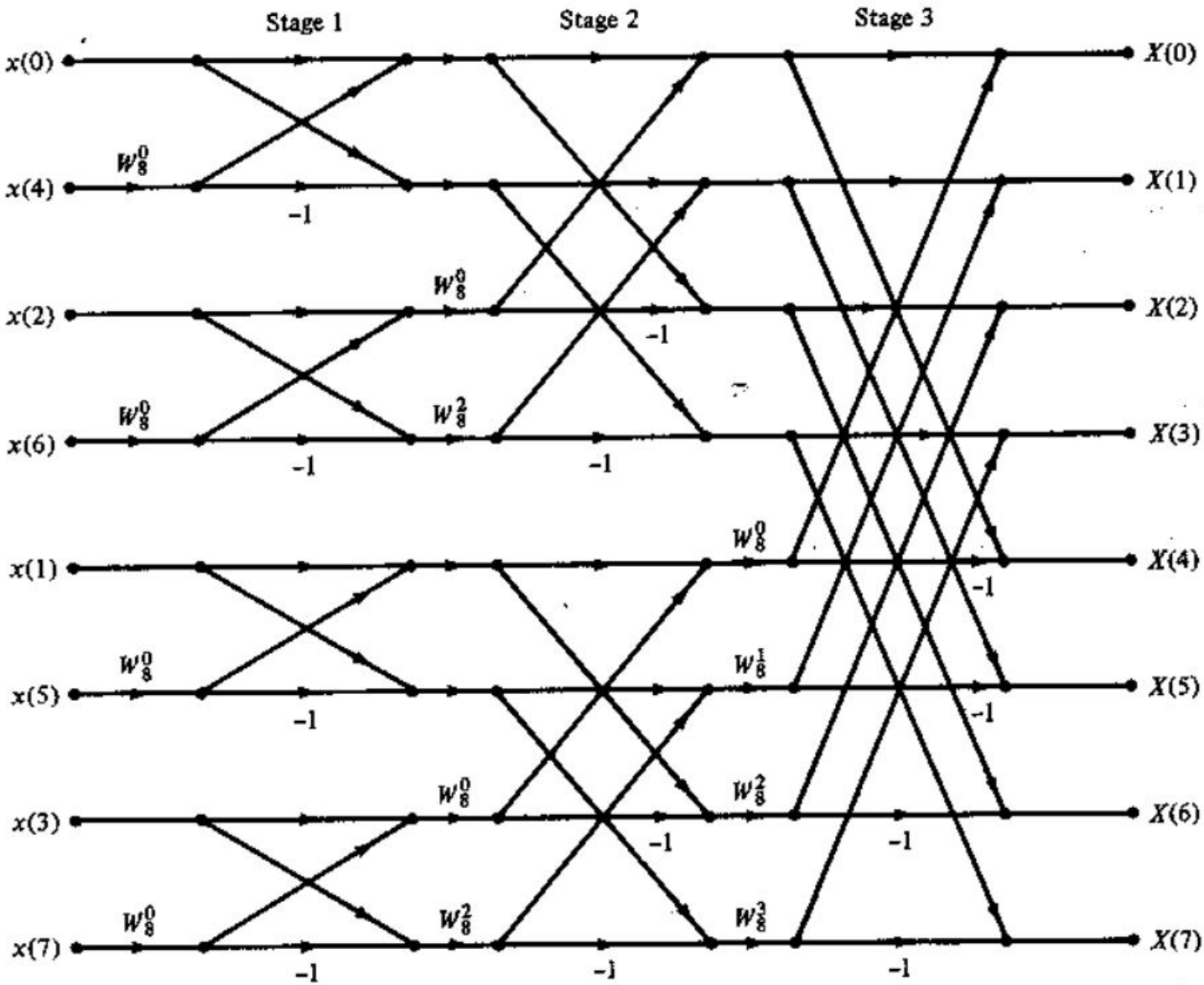


Figure Eight-point decimation-in-time FFT algorithm.

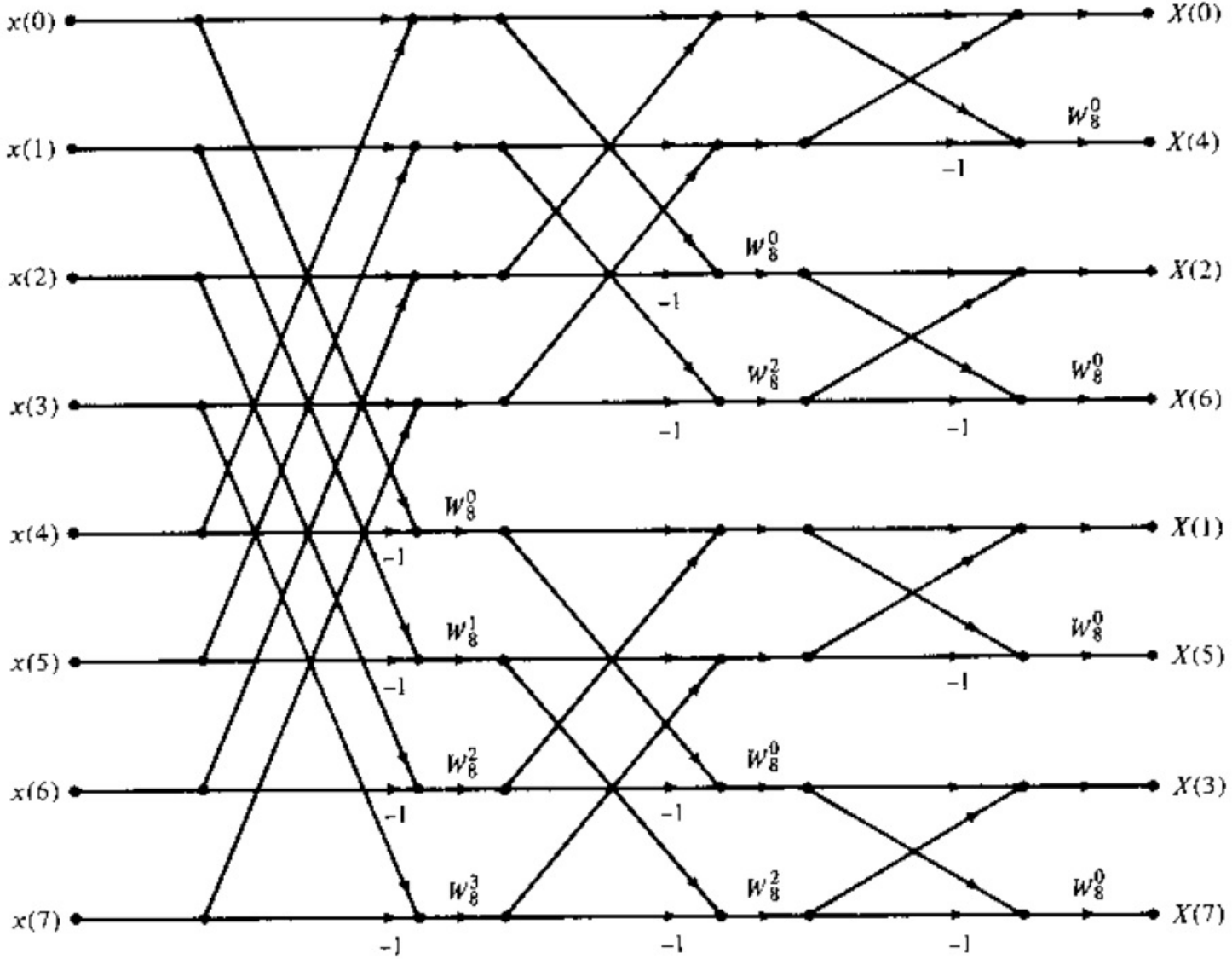


Figure $N = 8$-piont decimation-in-frequency FFT algorithm.

$x(n) = \{1, 1, 1, 0, 0, 0, 0, 0\}$

$X(K)$ using DIT-FFT algorithm.

(I) $x(0), x(4), x(2), x(6), x(1), x(5), x(3), x(7)$

$1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0$

O/P of Stage 1

$[1, 1, 1, -j, 1, 1, 0, 0]$

O/P of Stage 2

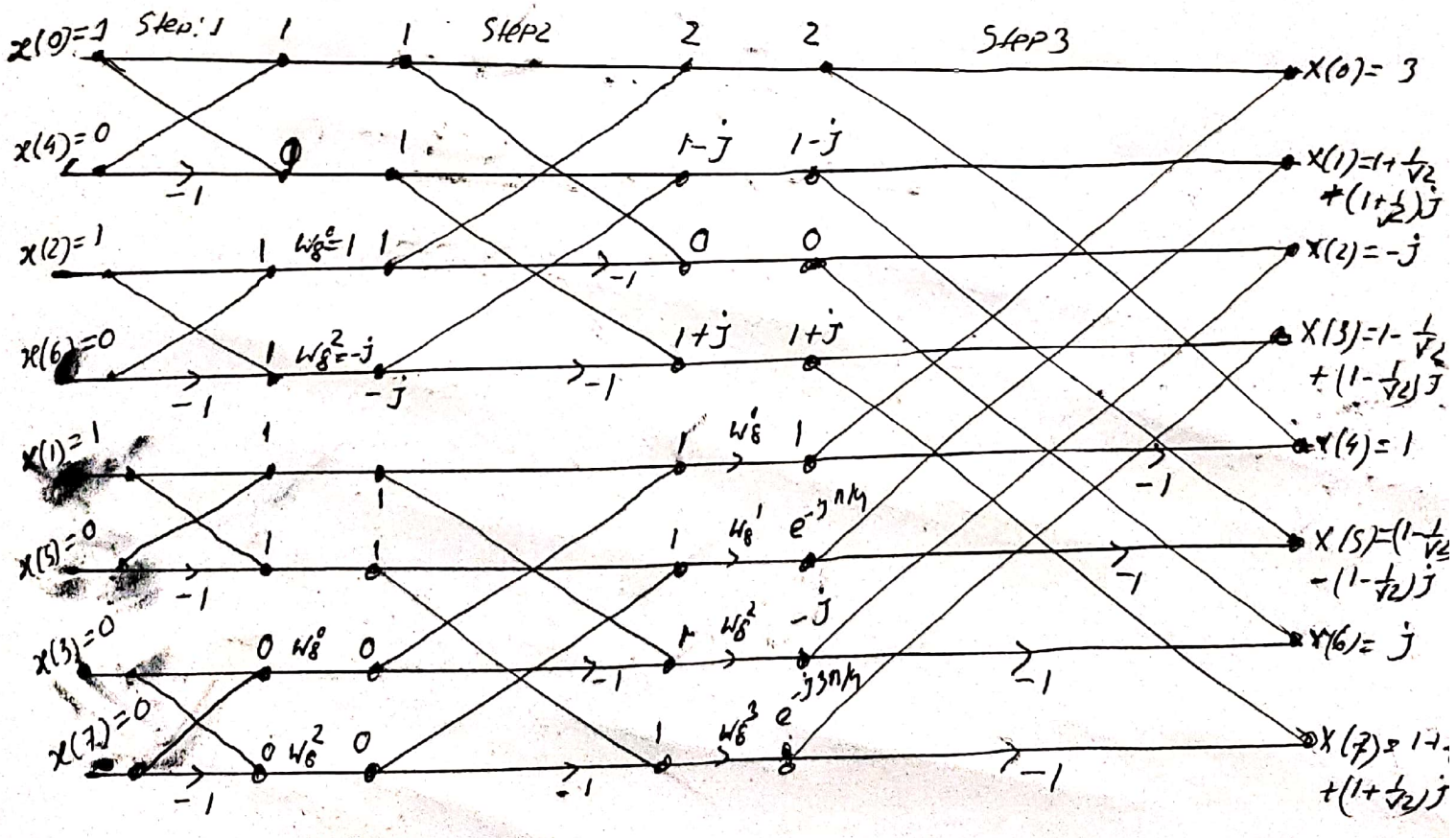
$[2, 1-j, 0, 1+j, 1, e^{-j\pi/4}, -j, e^{-j3\pi/4}]$

O/P of Stage 3

$X(0)$	$X(1)$	$X(2)$	$X(3)$	$X(4)$	$X(5)$
3	$1+\frac{1}{\sqrt{2}}-(1+\frac{1}{\sqrt{2}})j$	$-j$	$1-\frac{1}{\sqrt{2}}+(1-\frac{1}{\sqrt{2}})j$	1	$(1-\frac{1}{\sqrt{2}})-(1-\frac{1}{\sqrt{2}})j$

$X(6)$	$X(7)$
j	$1+\frac{1}{\sqrt{2}}+(1+\frac{1}{\sqrt{2}})j$

$[2, 1-j, 0, 1+j, 1, e^{-j\pi/4}, -j, e^{-j3\pi/4}]$



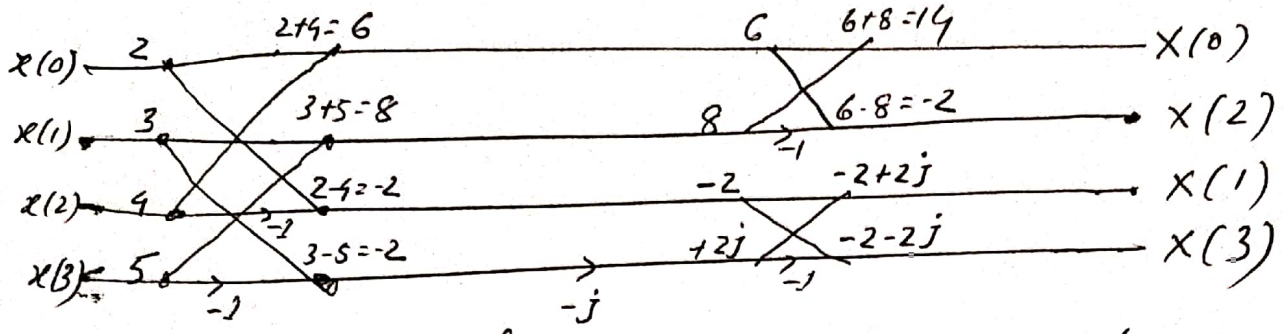
$W_8^2 = (e^{-j2\pi/8})^2 = e^{-j\pi\pi/8} = e^{-j\pi/2} = -j$

$W_8^1 = e^{-j2\pi/8} = e^{-j\pi/4} = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j$

Scanned by CamScanner

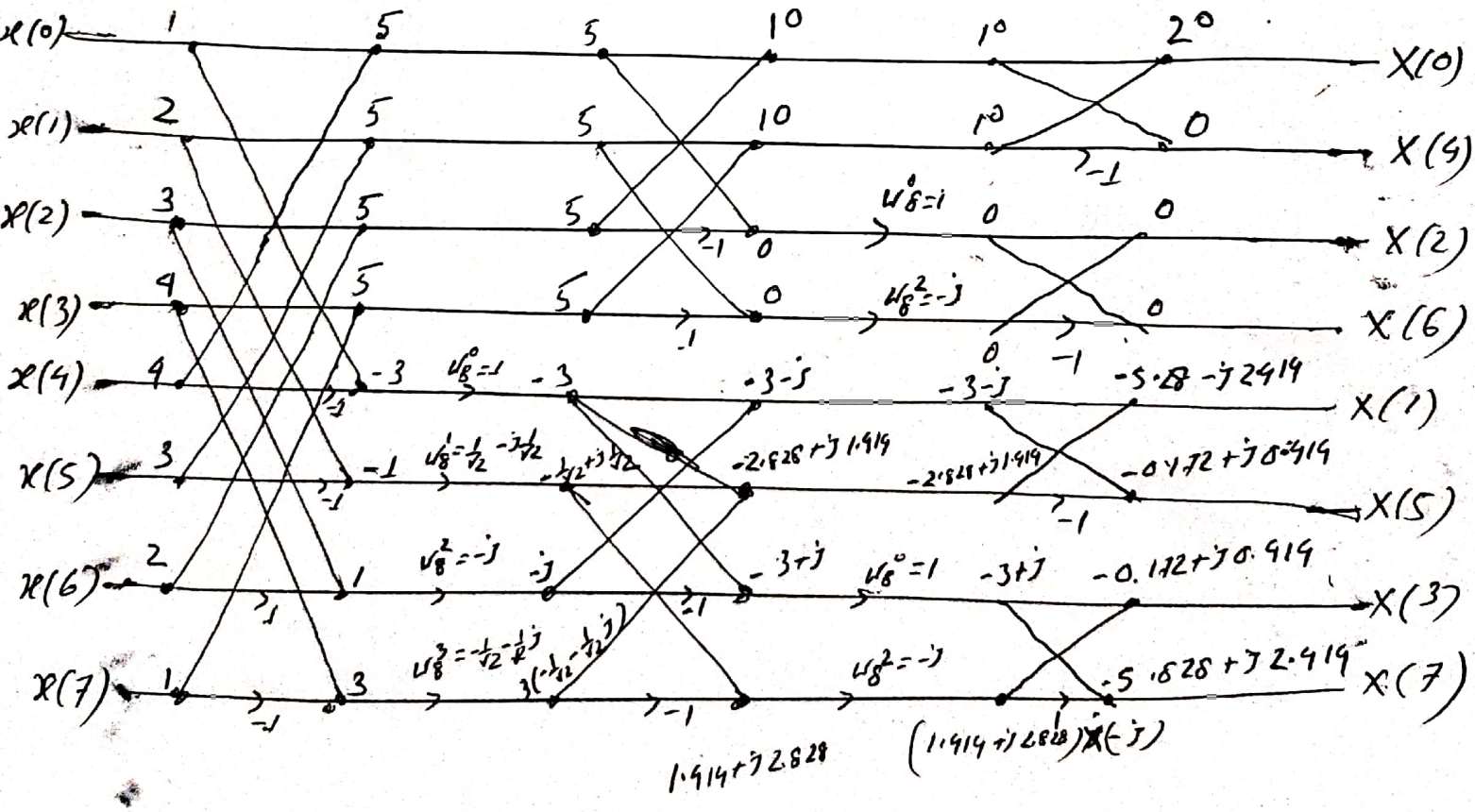
DIF-FFT

Q. Compute 4pt. DIF-FFT of given $x(n) = \{2, 3, 4, 5\}$



$$X(k) = \{14, -2+2j, -2, -2-2j\}$$

Q. Compute the DFT (i.e. FFT) for the sequence $\{1, 2, 3, 4, 4, 3, 2, 1\}$ using radix-2 DIF-FFT algorithm.



$$X(k) = \{20, -5.28-j2.414, \cancel{-0.172+j0.414}\ 0, -0.172+j0.414, 0, -0.172+j0.414, 0, -5.828+j2.414\}$$

Scanned by CamScanner